

DBMS – PAYMENT MANAGEMENT

Project Title: ONE8 Footwear E-Commerce System

Skill Task: DBMS

Week: 3

1. Database Design

1.1 Orders Table (prerequisite reference)

Attribute	Data Type	Constraint
Order_ID	NUMBER	Primary Key
Customer_ID	NUMBER	NOT NULL
Order_Date	DATE	—
Total_Amount	NUMBER(10,2)	—
Order_Status	VARCHAR2(20)	—

1.2 Payment Table

Attribute	Data Type	Constraint
Payment_ID	NUMBER	Primary Key
Order_ID	NUMBER	NOT NULL, Foreign Key
Payment_Mode	VARCHAR2(30)	NOT NULL
Payment_Date	DATE	NOT NULL
Amount	NUMBER(10,2)	NOT NULL
Payment_Status	VARCHAR2(20)	NOT NULL, DEFAULT 'Pending', CHECK IN ('Success','Failed','Pending')

2. Primary and Foreign Key Relationship

Orders → Payment

- Orders.Order_ID → Primary Key
- Payment.Payment_ID → Primary Key
- Payment.Order_ID → Foreign Key referencing Orders.Order_ID
- Relationship: One-to-One (1:1) — every order has exactly one payment record

3. Table Creation

3.1 Create Orders Table

```
SQL> CREATE TABLE Orders (  
 2     Order_ID      NUMBER PRIMARY KEY,  
 3     Customer_ID   NUMBER NOT NULL,  
 4     Order_Date    DATE,  
 5     Total_Amount  NUMBER(10,2),  
 6     Order_Status  VARCHAR2(20)  
 7 );
```

Table created.

3.2 Create Payment Table

```
SQL> CREATE TABLE Payment (  
 2     Payment_ID    NUMBER PRIMARY KEY,  
 3     Order_ID       NUMBER NOT NULL,  
 4     Payment_Mode   VARCHAR2(30) NOT NULL,  
 5     Payment_Date   DATE NOT NULL,  
 6     Amount         NUMBER(10,2) NOT NULL,  
 7     Payment_Status VARCHAR2(20) DEFAULT 'Pending' NOT NULL,  
 8     CONSTRAINT FK_Payment_Order  
 9         FOREIGN KEY (Order_ID)  
10         REFERENCES Orders(Order_ID),  
11     CONSTRAINT CHK_Payment_Status  
12         CHECK (Payment_Status IN ('Success', 'Failed', 'Pending'))  
13 );
```

Table created.

4. Data Insertion

4.1 Orders Records

```
SQL> INSERT INTO Orders VALUES (201, 1, TO_DATE('05-JAN-26','DD-MON-YY'), 3299.00, 'Delivered');

1 row created.

SQL>
SQL> INSERT INTO Orders VALUES (202, 2, TO_DATE('06-JAN-26','DD-MON-YY'), 4199.00, 'Delivered');

1 row created.

SQL>
SQL> INSERT INTO Orders VALUES (203, 3, TO_DATE('08-JAN-26','DD-MON-YY'), 1999.00, 'Shipped');

1 row created.

SQL>
SQL> INSERT INTO Orders VALUES (204, 4, TO_DATE('09-JAN-26','DD-MON-YY'), 1499.00, 'Delivered');

1 row created.

SQL>
SQL> INSERT INTO Orders VALUES (205, 5, TO_DATE('10-JAN-26','DD-MON-YY'), 4499.00, 'Pending');

1 row created.

SQL>
SQL> INSERT INTO Orders VALUES (206, 6, TO_DATE('11-JAN-26','DD-MON-YY'), 4999.00, 'Delivered');

1 row created.

SQL>
SQL> INSERT INTO Orders VALUES (207, 7, TO_DATE('12-JAN-26','DD-MON-YY'), 5499.00, 'Cancelled');

1 row created.

SQL>
SQL> INSERT INTO Orders VALUES (208, 8, TO_DATE('13-JAN-26','DD-MON-YY'), 5999.00, 'Shipped');

1 row created.
```

4.2 Payment Records

```
SQL> INSERT INTO Payment VALUES (301, 201, 'UPI', TO_DATE('05-JAN-26','DD-MON-YY'), 3299.00, 'Success');

1 row created.

SQL>
SQL> INSERT INTO Payment VALUES (302, 202, 'Credit Card', TO_DATE('06-JAN-26','DD-MON-YY'), 4199.00, 'Success');

1 row created.

SQL>
SQL> INSERT INTO Payment VALUES (303, 203, 'Net Banking', TO_DATE('08-JAN-26','DD-MON-YY'), 1999.00, 'Success');

1 row created.

SQL>
SQL> INSERT INTO Payment VALUES (304, 204, 'Debit Card', TO_DATE('09-JAN-26','DD-MON-YY'), 1499.00, 'Failed');

1 row created.
```

```
SQL> INSERT INTO Payment VALUES (305, 205, 'UPI', TO_DATE('10-JAN-26','DD-MON-YY'), 4499.00, 'Pending');

1 row created.

SQL>
SQL> INSERT INTO Payment VALUES (306, 206, 'Credit Card', TO_DATE('11-JAN-26','DD-MON-YY'), 4999.00, 'Success');

1 row created.

SQL>
SQL> INSERT INTO Payment VALUES (307, 207, 'Cash on Delivery', TO_DATE('12-JAN-26','DD-MON-YY'), 5499.00, 'Failed');

1 row created.

SQL>
SQL> INSERT INTO Payment VALUES (308, 208, 'UPI', TO_DATE('13-JAN-26','DD-MON-YY'), 5999.00, 'Success');

1 row created.
```

5. Data Retrieval

5.1 Display Payment Records

```
SQL> SELECT * FROM Payment;
```

PAYMENT_ID	ORDER_ID	PAYMENT_MODE	PAYMENT_D	AMOUNT	PAYMENT_STATUS
301	201	UPI	05-JAN-26	3299.00	Success
302	202	Credit Card	06-JAN-26	4199.00	Success
303	203	Net Banking	08-JAN-26	1999.00	Success
304	204	Debit Card	09-JAN-26	1499.00	Failed
305	205	UPI	10-JAN-26	4499.00	Pending
306	206	Credit Card	11-JAN-26	4999.00	Success
307	207	Cash on Delivery	12-JAN-26	5499.00	Failed
308	208	UPI	13-JAN-26	5999.00	Success

8 rows selected.

6. Managing Successful and Failed Transactions

View all successful transactions:

```
SQL> SELECT * FROM Payment
2 WHERE Payment_Status = 'Success';
```

PAYMENT_ID	ORDER_ID	PAYMENT_MODE	PAYMENT_D	AMOUNT	PAYMENT_STATUS
301	201	UPI	05-JAN-26	3299.00	Success
302	202	Credit Card	06-JAN-26	4199.00	Success
303	203	Net Banking	08-JAN-26	1999.00	Success
306	206	Credit Card	11-JAN-26	4999.00	Success
308	208	UPI	13-JAN-26	5999.00	Success

5 rows selected.

View all failed transactions:

```
SQL> SELECT * FROM Payment
2 WHERE Payment_Status = 'Failed';
```

PAYMENT_ID	ORDER_ID	PAYMENT_MODE	PAYMENT_D	AMOUNT	PAYMENT_STATUS
304	204	Debit Card	09-JAN-26	1499.00	Failed
307	207	Cash on Delivery	12-JAN-26	5499.00	Failed

2 rows selected.

Retry a failed payment and mark it successful:

```
SQL> UPDATE Payment
2 SET Payment_Status = 'Success',
3     Payment_Date = TO_DATE('14-01-2026', 'DD-MM-
4 WHERE Payment_ID = 304;
```

1 row updated.

Resolve a pending payment:

```
SQL> UPDATE Payment
2 SET Payment_Status = 'Success'
3 WHERE Payment_ID = 305;
```

1 row updated.

Verify the updated records:

```
SQL> SELECT * FROM Payment
2 WHERE Payment_ID IN (304, 305);
```

PAYMENT_ID	ORDER_ID	PAYMENT_MODE	PAYMENT_D	AMOUNT	PAYMENT_STATUS
304	204	Debit Card	14-JAN-26	1499.00	Success
305	205	UPI	10-JAN-26	4499.00	Success

2 rows selected.

7. Payment Method Analysis

Total transactions and amount collected by payment mode:

```

SQL> SELECT
  2     Payment_Mode,
  3     COUNT(*) AS Total_Transactions,
  4     SUM(Amount) AS Total_Amount
  5 FROM Payment
  6 GROUP BY Payment_Mode
  7 ORDER BY Total_Transactions DESC;

```

PAYMENT_MODE	TOTAL_TRANSACTIONS	TOTAL_AMOUNT
UPI	3	13797.00
Credit Card	2	9198.00
Cash on Delivery	1	5499.00
Debit Card	1	1499.00
Net Banking	1	1999.00

5 rows selected.

Payment status summary:

```

SQL> SELECT
  2     Payment_Status,
  3     COUNT(*) AS Total_Transactions
  4 FROM Payment
  5 GROUP BY Payment_Status
  6 ORDER BY Payment_Status;

```

PAYMENT_STATUS	TOTAL_TRANSACTIONS
Failed	1
Success	7

2 rows selected.

8. Payment Transaction Reports

This report displays each payment transaction along with the customer name, order, payment mode, date, amount and status, ordered by payment date.

```

SQL> SELECT
  2     c.Customer_Name,
  3     o.Order_ID,
  4     p.Payment_Mode,
  5     p.Payment_Date,
  6     p.Amount,
  7     p.Payment_Status
  8 FROM Payment p
  9 JOIN Orders o
 10     ON p.Order_ID = o.Order_ID
 11 JOIN Customer c
 12     ON o.Customer_ID = c.Customer_ID
 13 ORDER BY p.Payment_Date;

```

CUSTOMER_NAME	ORDER_ID	PAYMENT_MODE	PAYMENT_D	AMOUNT	STATUS
Renu Srinivasan	201	UPI	05-JAN-26	3299.00	Success
Arjun Kumar	202	Credit Card	06-JAN-26	4199.00	Success
Priya Kumar	203	Net Banking	08-JAN-26	1999.00	Success
Keerthana Mohan	205	UPI	10-JAN-26	4499.00	Success
Nandhini Ravi	206	Credit Card	11-JAN-26	4999.00	Success
Karthik Prakash	207	Cash on Delivery	12-JAN-26	5499.00	Failed
Swetha Balaji	208	UPI	13-JAN-26	5999.00	Success
Rahul Sharma	204	Debit Card	14-JAN-26	1499.00	Success

8 rows selected.